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Reciprocity and sensitivity kernels for sea level fingerprints

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SUMMARY

Reciprocity theorems are established for the elastic sea level fingerprint problem including rotational feedbacks. In their simplest form, these results show that the sea level change at a location **x** due to melting a unit point mass of ice at **x**′ is equal to the sea level change at **x**′ due to melting a unit point mass of ice at **x**. This identity holds irrespective of the shoreline geometry or of lateral variations in elastic Earth structure. Using the reciprocity theorems, sensitivity kernels for sea level and related observables with respect to the ice load can be readily derived. It is notable that calculation of the sensitivity kernels is possible using standard fingerprint codes, though for some types of observable a slight generalization to the fingerprint problem must be considered. These results are of use within coastal hazard assessment and have a range of applications within studies of modern-day sea level change. To illustrate the latter point, we use sensitivity kernels to investigate two widely used methods for estimating, respectively, ice sheet mass loss from satellite gravity, and rates of global mean sea level rise from satellite altimetry. Though our analysis is idealized in some respects, we identify systematic errors of order 5 per cent due to the use of simplified sea level physics. Crucially, calculation of the relevant sensitivity kernels provides not only a means for understanding sources of bias in existing methods, but will aid in the design of new and improved data-assimilation techniques.

Key words: Loading of the Earth; Sea level change; Inverse theory.

1 INTRODUCTION

Farrell & Clark (1976) provided the first gravitationally self-consistent theory of post-glacial sea level change on a non-rotating, 1-D (depth varying) viscoelastic Earth for the case of time-invariant shoreline geometry. Their static 'sea level equation' has played a central role in the modern development of the field of glacial isostatic adjustment (GIA) and, in the intervening half century, their theoretical treatment has been extended to incorporate 3-D viscoelastic Earth structure and to include both rotational effects on sea level and shoreline migration due either to local sea level fluctuations or changes in the perimeter of grounded, marine-based ice (Mitrovica & Milne 2003; Kendall *et al.* 2005).

The predictions appearing in Farrell & Clark (1976) included the special case of elastic Earth models (e.g. their figs 3 and 4), appropriate for considering the sea level response to ice mass flux with a time scale shorter than the viscous relaxation time. The calculations highlighted the counter-intuitive result that sea level falls in the vicinity of a rapidly melting ice sheet due to the combined effects of elastic crustal uplift and the loss of gravitational attraction towards the diminishing ice cover. The maximum sea level fall adjacent to the ice sheet is an order of magnitude (or more) greater than the global mean sea level (GMSL) rise associated with the melt event, while at large distances from the ice sheet the predicted sea level rise is up to ∼30 per cent greater than GMSL.

Clark & Lingle (1977) and later Clark & Primus (1987) highlighted the relevance of the elastic case for predicting geographically variable sea level change arising from ice sheet mass flux driven by modern global warming. The latter was cited by Mercer (1978) in his canonical study of the vulnerability of the West Antarctic Ice Sheet to climate change and the implications of its potential collapse for the Earth system. The connection was further reinforced by Conrad & Hager (1997) who quantified the potential bias in estimates of GMSL based on the existing distribution of tide gauge records. Mitrovica *et al.* (2001) were the first to demonstrate that geographic variability in a subset of tide-gauge determined sea level rates could be reconciled by specific combinations of melt sources. Their predictions included rotational effects and they emphasized the unique geometry, or fingerprints (Plag & Juettner 2001), of sea level change associated with each ice sheet or glacier. The notion of 'fingerprinting' the individual sources of melt-water has subsequently become a central theme in the analysis of modern sea level observations (e.g. Tamisiea *et al.* 2001; Milne *et al.* 2009; Plag 2006; Bamber & Riva 2010; Mitrovica *et al.* 2011;

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Brunnabend *et al.* 2015; Spada & Galassi 2016). Moreover, the inclusion of fingerprint physics in such analysis has led to a significant revision in estimates of 20th century GMSL (Hay *et al.* 2015; Dangendorf *et al.* 2017).

In all the above analyses, sea level fingerprints were predicted for specific, assumed geometries of ice mass flux and this has hindered the adoption of such models in assessments of coastal hazards posed by a warming world. Three recent studies have inverted these analyses by developing methods for producing a map that shows the variation in sensitivity of sea level at a specific site to ice mass changes anywhere within the cryosphere (Larour *et al.* 2017; Mitrovica *et al.* 2018; Crawford *et al.* 2018). These maps represent sensitivity kernels that can be integrated with arbitrary ice mass flux geometries to compute the sea level changes at any site of interest. The Larour *et al.* (2017) approach was based on an automatic differentiation tool applied to sea level software that solved the forward problem (i.e. computation of the sea level fingerprint associated with a specific ice mass flux), while Mitrovica *et al*. (2018) solved a large number of forward problems in which a grid-by-grid perturbation in ice thickness was prescribed and applied a finite-difference approximation. Finally, Crawford *et al.* (2018) applied the adjoint method to the equations governing the full sea level problem within a viscoelastic earth model, obtaining results for elastic fingerprints as a special case. An advantage of this latter method is that it yields exact sensitivity kernels at a cost equivalent to just two forward calculations (and just one in the case of the fingerprint problem). A limitation of that study, along with earlier work on adjoint methods for GIA (Al-Attar & Tromp 2014; Martinec *et al.* 2015), was the neglect of rotational feedbacks. Moreover, the results of Crawford *et al.* (2018) that are relevant specifically to fingerprints are embedded within the more complex viscoelastic theory which makes them harder to understand and apply.

In this paper, we revisit the derivation of sensitivity kernels through a different and simpler argument that is specialized to the elastic fingerprint problem with rotational feedbacks included and with a fixed shoreline geometry. The latter assumption renders the fingerprint problem linear and is appropriate within applications to modern sea level where amplitudes of change are low. Our presentation is deliberately independent of Crawford *et al.* (2018), and hence a knowledge of that work and the Lagrange multiplier methods on which it is based is not necessary to understand this paper. We proceed here by first establishing reciprocity theorems for the fingerprint problem along with suitable generalizations thereof. In their simplest form, these results show the following: Suppose that at some location, $\mathbf{x}$, a unit point mass of ice is melted and the sea level change is observed at $\mathbf{x}'$. This sea level change is identical to that that would be observed at $\mathbf{x}$ due to melting a unit point mass of ice at $\mathbf{x}'$. The derivation makes no assumptions about the shoreline geometry nor internal structure of the earth model, and hence this is not merely a geometric symmetry of the problem. With the reciprocity theorems at hand, it is a simple matter to obtain sensitivity kernels for the whole range of physically relevant observables. In the case of observables depending directly on sea level change (e.g. tide gauge determined rates), calculation of the sensitivity kernel can be performed using a standard fingerprint code. When more general observables are considered (e.g. those associated with satellite altimetry or gravity) a generalized form of the fingerprint problem must be solved, but the necessary changes are slight. While the focus of this work is on sea level change driven by the growth or melting of continental ice sheets, it is worth commenting that the theory developed is applicable to other types of surface loading such as that associated with hydrology, ocean dynamics or sedimentation.

2 SENSITIVITY KERNELS IN NON-ROTATING EARTH MODELS

Within this section we derive reciprocity theorems for the elastic fingerprint problem in a non-rotating earth model and show how they can be used to obtain sensitivity kernels for sea level (and other observables) with respect to ice thickness. The necessary extensions to account for rotational feedbacks are described in Section 3. Splitting the theory in this manner is unnecessary but we think it helpful to incrementally add complicating factors into the problem.

2.1 Equations of motion for static loading

We start by considering the static loading of a non-rotating elastic earth model. The volume of the earth model will be written M and its surface ∂M. It is assumed that, prior to application of the load, the earth model is in a state of hydrostatic equilibrium, with its gravitational potential written Φ. When subject to a load, σ, the earth model will deform, with the displacement vector written, $\mathbf{u}$ and the perturbed gravitational potential, ϕ. The equations of motion appropriate for this situation are well-known (e.g. Dahlen 1974; Dahlen & Tromp 1998; Tromp & Mitrovica 1999). For our purposes, it is useful to start from the weak-formulation of the equations of motion derived by Al-Attar & Tromp (2014, section 2.3.2). Within this approach, all equations of motion and boundary conditions can be concisely stated as

$$\mathcal{A}(\mathbf{u}, \phi \,|\, \mathbf{u}', \phi') + \int_{\partial M} (\mathbf{u}' \cdot \nabla\Phi + \phi')\,\sigma \,\mathrm{d}S = 0. \tag{1}$$

Here $\mathbf{u}'$ and ϕ' are test functions for the displacement and gravitational potential, respectively, and the stated equality is required to hold for all possible values of these functions. The term $\mathcal{A}(\mathbf{u}, \phi \,|\, \mathbf{u}', \phi')$, which is associated with the elastic and gravitational response of the earth model, is a bilinear form whose definition can be found in eq. (2.52) of Al-Attar & Tromp (2014). In what follows, we need only two properties of this bilinear form:

(i) It is symmetric in the sense that

$$\mathcal{A}(\mathbf{u}, \phi \,|\, \mathbf{u}', \phi') = \mathcal{A}(\mathbf{u}', \phi' \,|\, \mathbf{u}, \phi). \tag{2}$$

Note that within the identity the displacement and gravitational potential perturbations are paired together; we cannot, for example, interchange the two displacements vectors alone.

(ii) For any test functions $(\mathbf{u}', \phi')$, the following identity holds

$$\mathcal{A}(\mathbf{u}, \phi \,|\, \mathbf{u}', \phi') = 0 \tag{3}$$

when the displacement and gravitational potential perturbation take the form

$$\mathbf{u} = \mathbf{a} + \mathbf{b} \times \mathbf{x}, \quad \phi = -(\mathbf{a} + \mathbf{b} \times \mathbf{x}) \cdot \nabla\Phi, \tag{4}$$

with arbitrary constant vectors $\mathbf{a}$ and $\mathbf{b}$. The fields $\mathbf{u}$ and ϕ defined here correspond to a linearized rigid body motion, with the vector $\mathbf{a}$ parametrizing the translational degrees of freedom and $\mathbf{b}$ those of rotation.

Using these two properties, we can take

$$\mathbf{u}' = \mathbf{a} + \mathbf{b} \times \mathbf{x}, \quad \phi' = -(\mathbf{a} + \mathbf{b} \times \mathbf{x}) \cdot \nabla\Phi, \tag{5}$$

within eq. (1) to arrive at a trivial equality. From this it follows via the Fredholm alternative (e.g. Marsden & Hughes 1983, chapter 6) that the static loading problem admits a solution for any σ, but that the solution is only defined up to a linearized rigid body motion (*cf.* Martinec *et al.* 2015, Section 3).

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2.2 A first reciprocity theorem

Suppose that $(\mathbf{u}, \phi)$ solve the static loading problem for a load σ, and that $(\mathbf{u}^\dagger, \phi^\dagger)$ solve the static loading problem for some other load $\sigma^\dagger$. From eq. (1), we know that

$$\mathcal{A}(\mathbf{u}, \phi \,|\, \mathbf{u}^\dagger, \phi^\dagger) + \int_{\partial M} (\mathbf{u}^\dagger \cdot \nabla\Phi + \phi^\dagger)\,\sigma \,\mathrm{d}S = 0. \tag{6}$$

In exactly the same way for the second problem, we can write

$$\mathcal{A}(\mathbf{u}^\dagger, \phi^\dagger \,|\, \mathbf{u}, \phi) + \int_{\partial M} (\mathbf{u} \cdot \nabla\Phi + \phi)\,\sigma^\dagger \,\mathrm{d}S = 0. \tag{7}$$

Subtracting one equality from the other and using the symmetry of $\mathcal{A}$, we arrive at the non-trivial identity

$$\int_{\partial M} (\mathbf{u}^\dagger \cdot \nabla\Phi + \phi^\dagger)\,\sigma \,\mathrm{d}S = \int_{\partial M} (\mathbf{u} \cdot \nabla\Phi + \phi)\,\sigma^\dagger \,\mathrm{d}S. \tag{8}$$

This is not a new result. It was obtained by Tromp & Mitrovica (1999) within the context of quasi-static loading problems and is, as they pointed out, a generalization of Betti's reciprocity theorem that is well-known in seismology (e.g. Aki & Richards 2002).

2.3 Reformulating the theorem in terms of sea level

Within gravitationally self-consistent sea level theory (e.g. Farrell & Clark 1976; Mitrovica & Milne 2003; Crawford *et al.* 2018), deformation of the earth model can be related to sea level change, ΔSL, through the relation

$$\Delta SL = -\frac{1}{g}(\mathbf{u} \cdot \nabla\Phi + \phi) + \frac{\Phi_g}{g}, \tag{9}$$

where g is the magnitude of gravitational acceleration at the surface. Here we note that we are working with a hydrostatic theory in which sea level is defined as difference been the equipotential that defines the sea surface and the solid surface (e.g. Mitrovica & Milne 2003; Tamisiea 2011). The first term on the right-hand side of eq. (9) is associated with vertical motion of the solid surface, the second with changes in the shape of equipotential surfaces, while the spatially constant final term, $\frac{\Phi_g}{g}$, is linked to variations in ocean mass. As a comment on notations, within this paper, we consider sea level change, ΔSL, along with related quantities such as ice thickness change, or global mean sea level change that are defined later. Because, however, all equations are linear and have no explicit time-dependence, we could equally well have chosen to work with with rates of change of these quantities. The value of the constant Φ_g is fixed by conserving mass between the oceans and direct load, with this requirement conveniently written as

$$\int_{\partial M} \sigma \,\mathrm{d}S = 0. \tag{10}$$

Here the total load, σ, is decomposed into an ocean load and a direct load, ζ, as

$$\sigma = \rho_\mathrm{w} C\,\Delta SL + \zeta, \tag{11}$$

where ρ_w is the density of water and C the ocean function that equals one where water is present and zero otherwise. In the case of ice loading, the direct term is given by

$$\zeta = \rho_\mathrm{i}(1 - C)\Delta I, \tag{12}$$

where ρ_i is the density of ice, and ΔI the change in ice thickness. The factor, $1 - C$, within this expression accounts for the possibility of floating ice (e.g. Crawford *et al.* 2018, eqs 31–34).

Consider again a pair of solutions $(\mathbf{u}, \phi)$ and $(\mathbf{u}^\dagger, \phi^\dagger)$ of the loading problem associated, respectively, to with loads σ and $\sigma^\dagger$. Here, however, we assume that these loads are decomposed as in eq. (11) into water and direct terms that share a common ocean function. From the above expression for sea level change we can write

$$\mathbf{u} \cdot \nabla\Phi + \phi = -g\,\Delta SL + \Phi_g, \quad \mathbf{u}^\dagger \cdot \nabla\Phi + \phi^\dagger = -g\,\Delta SL^\dagger + \Phi_g^\dagger, \tag{13}$$

and hence eq. (8) becomes

$$\int_{\partial M} \left(-g\,\Delta SL^\dagger + \Phi_g^\dagger\right)\,\sigma \,\mathrm{d}S = \int_{\partial M} (-g\,\Delta SL + \Phi_g)\,\sigma^\dagger \,\mathrm{d}S. \tag{14}$$

The terms involving the constants Φ_g and $\Phi_g^\dagger$ vanish due to conservation of mass from eq. (10), and so the identity simplifies to

$$\int_{\partial M} \Delta SL^\dagger\,\sigma \,\mathrm{d}S = \int_{\partial M} \Delta SL\,\sigma^\dagger \,\mathrm{d}S. \tag{15}$$

If we now substitute into the equality the decompositions of the loads, σ and $\sigma^\dagger$, we find

$$\int_{\partial M} \Delta SL^\dagger(\rho_\mathrm{w} C\,\Delta SL + \zeta)\,\mathrm{d}S = \int_{\partial M} \Delta SL\,(\rho_\mathrm{w} C\,\Delta SL^\dagger + \zeta^\dagger)\,\mathrm{d}S, \tag{16}$$

and cancelling the terms symmetric in ΔSL and $\Delta SL^\dagger$ we arrive at

$$\int_{\partial M} \Delta SL^\dagger\,\zeta \,\mathrm{d}S = \int_{\partial M} \Delta SL\,\zeta^\dagger \,\mathrm{d}S. \tag{17}$$

Again, this is a known result, being a special case by the adjoint theory of Crawford *et al.* (2018) for sea level change in a viscoelastic earth model. What is new is the explicit statement as a reciprocity theorem along with the more elementary derivation that has been facilitated by restricting attention to the elastic fingerprint problem.

2.4 Symmetry of the Green's function

Because the fingerprint problem is linear, its solution must take the form

$$\Delta SL(\mathbf{x}) = \int_{\partial M} G(\mathbf{x}, \mathbf{x}')\zeta(\mathbf{x}')\,\mathrm{d}S', \tag{18}$$

for an appropriate Green's function, and where the integral is taken with respect to $\mathbf{x}'$. Suppose that, within eq. (17), we take

$$\zeta^\dagger(\mathbf{x}') = \delta(\mathbf{x}', \mathbf{x}), \tag{19}$$

with $\delta(\mathbf{x}', \mathbf{x})$ being the delta function on ∂M based at $\mathbf{x}$ which is defined such that

$$\int_{\partial M} f(\mathbf{x}')\delta(\mathbf{x}', \mathbf{x})\,\mathrm{d}S' = f(\mathbf{x}), \tag{20}$$

for any suitably regular function, f. By definition of the Green's function, it follows that $\Delta SL^\dagger(\mathbf{x}') = G(\mathbf{x}', \mathbf{x})$, and hence from the reciprocity theorem we arrive at the identity

$$G(\mathbf{x}, \mathbf{x}') = G(\mathbf{x}', \mathbf{x}). \tag{21}$$

Such a symmetry of the Green's function was established for static loading problems by Tromp & Mitrovica (1999) in cases where there is only a direct load. Here we see the result remains true when a gravitationally self-consistent water load is included. It is this symmetry that makes precise our earlier statements of the reciprocity theorem in terms of the response to melting unit point masses of ice.

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2.5 Self-adjointness of the sea level equation

An alternative description of the sea level reciprocity theorem can be given in the language of functional analysis. Solution of the sea level equation implicitly defines a linear mapping, A, from the direct load, ζ, to the resulting sea level change, ΔSL, with both scalar fields being defined on the surface, ∂M, of the earth model. Using this notation within eq. (17), we have

$$\int_{\partial M} (A\zeta^\dagger)\,\zeta \,\mathrm{d}S = \int_{\partial M} (A\zeta)\,\zeta^\dagger \,\mathrm{d}S. \tag{22}$$

Regarding the integral, $\int_{\partial M} uv \,\mathrm{d}S$, of two real-valued scalar fields, u and v, on ∂M as an inner product, it follows that the operator, A, is self-adjoint (e.g. Schechter 2001). To make these ideas precise, one should specify the domain and range of the operator, A. We will not dwell on such technical points nor pursue this operator formalism further within this work, but note that these ideas can also be applied to the generalized reciprocity theorems discussed below.

2.6 Sensitivity kernels for sea level change

Suppose now that $J(\Delta SL)$ is some functional of the sea level change. For example, within the context of an inverse problem to recover the direct load we might consider

$$J(\Delta SL) = \frac{1}{2}\int_{\partial M} C\,(\Delta SL - \Delta SL_\mathrm{o})^2 \,\mathrm{d}S, \tag{23}$$

where ΔSL_o is the observed sea level change and the ocean function is included to limit contributions to the oceans. Because ΔSL is obtained by solving the sea level problem for a given direct load, ζ, this functional implicitly depends on ζ. It can, for a variety of reasons, be interesting to determine the functional derivative of J with respect to ζ. To do so, we write $\delta\zeta$ for the perturbation in the direct load, and $\delta\Delta SL$ for the corresponding perturbation to ΔSL. Working to first-order in perturbed quantities, we have

$$\delta J = \int_{\partial M} C\,(\Delta SL - \Delta SL_\mathrm{o})\,\delta\Delta SL \,\mathrm{d}S. \tag{24}$$

On the right-hand side we see $\delta\Delta SL$ multiplied by a weighting function, $C\,(\Delta SL - \Delta SL_\mathrm{o})$, determined by the form of J. For a general functional, $J(\Delta SL)$, we can similarly write

$$\delta J = \int_{\partial M} \zeta^\dagger\,\delta\Delta SL \,\mathrm{d}S, \tag{25}$$

where we write $\zeta^\dagger$ for the appropriate weighting function. Because the fingerprint problem is linear, eq. (17) immediately holds with $(\Delta SL, \zeta)$ replaced by $(\delta\Delta SL, \delta\zeta)$, and hence the reciprocity theorem then gives

$$\delta J = \int_{\partial M} \Delta SL^\dagger\,\delta\zeta \,\mathrm{d}S, \tag{26}$$

where $\Delta SL^\dagger$ is obtained by solving the fingerprint problem subject to the direct load $\zeta^\dagger$. This shows that $\Delta SL^\dagger$ is the sensitivity kernel of J with respect to ζ. Recalling that in the case of an ice load we have $\zeta = \rho_i(1-C)\Delta I$, it follows that we can equivalently write

$$\delta J = \int_{\partial M} \rho_\mathrm{i}\,(1-C)\,\Delta SL^\dagger\,\delta\Delta I \,\mathrm{d}S, \tag{27}$$

from which we can identify the sensitivity kernel of J with respect to the ice thickness

$$K = \rho_\mathrm{i}\,(1-C)\,\Delta SL^\dagger. \tag{28}$$

These results are special cases of those obtained by Crawford *et al.* (2018). Following the terminology of that work, we will call $\Delta SL^\dagger$ the adjoint sea level and $\zeta^\dagger$ the adjoint load. What this new derivation makes clear is that sensitivity kernels can be obtained by solving the standard fingerprint problem subject to an appropriate adjoint load.

2.7 Dealing with more complex observables

We have now seen that sensitivity kernels can be readily obtained for functionals, J, defined in terms of the sea level change, ΔSL. In practice there are, of course, other quantities that can be observed, including changes in sea surface height or in the gravitational potential. To obtain analogous results for such observables, it is necessary to consider the generalized loading problem

$$\mathcal{A}(\mathbf{u},\phi\,|\,\mathbf{u}',\phi') + \int_{\partial M} (\mathbf{u}'\cdot\nabla\Phi + \phi')\,\sigma \,\mathrm{d}S + \int_{\partial M} (\mathbf{u}'\cdot\mathbf{t} + \zeta_\phi\phi')\,\mathrm{d}S = 0, \tag{29}$$

where we have introduced additional forcing terms $\mathbf{t}$ and ζ_ϕ. These terms need not have a direct physical interpretation, but we will see that the term $\mathbf{t}$ allows us to consider functionals depending directly on the surface displacement, while ζ_ϕ is needed for those involving the gravitational potential perturbation. Recalling the second property of the bilinear form mentioned above, it follows that a necessary and sufficient condition for solutions to exist is

$$\int_{\partial M} (\mathbf{t} - \zeta_\phi\nabla\Phi)\cdot(\mathbf{a} + \mathbf{b}\times\mathbf{x})\,\mathrm{d}S = 0, \tag{30}$$

with $\mathbf{a}$ and $\mathbf{b}$ arbitrary constant vectors. The practical significance of this condition will be explained below.

A reciprocity theorem for the generalized loading problem can be obtained as before by introducing pairs of solutions $(\mathbf{u}, \phi)$ and $(\mathbf{u}^\dagger, \phi^\dagger)$ corresponding, respectively, to the generalized loads $(\sigma, \mathbf{t}, \zeta_\phi)$ and $(\sigma^\dagger, \mathbf{t}^\dagger, \zeta_\phi^\dagger)$. The result is the following identity

$$\int_{\partial M} (\mathbf{u}^\dagger\cdot\nabla\Phi + \phi^\dagger)\,\sigma \,\mathrm{d}S + \int_{\partial M} (\mathbf{t}\cdot\mathbf{u}^\dagger + \zeta_\phi\phi^\dagger)\,\mathrm{d}S = \int_{\partial M} (\mathbf{u}\cdot\nabla\Phi + \phi)\,\sigma^\dagger \,\mathrm{d}S + \int_{\partial M} (\mathbf{t}^\dagger\cdot\mathbf{u} + \zeta_\phi^\dagger\phi)\,\mathrm{d}S. \tag{31}$$

By decomposing the loads σ and $\sigma^\dagger$ into direct terms and matching water loads, the above identity becomes

$$\int_{\partial M} \Delta SL^\dagger\,\zeta \,\mathrm{d}S - \frac{1}{g}\int_{\partial M} (\mathbf{t}\cdot\mathbf{u}^\dagger + \zeta_\phi\phi^\dagger)\,\mathrm{d}S = \int_{\partial M} \Delta SL\,\zeta^\dagger \,\mathrm{d}S - \frac{1}{g}\int_{\partial M} (\mathbf{t}^\dagger\cdot\mathbf{u} + \zeta_\phi^\dagger\phi)\,\mathrm{d}S, \tag{32}$$

which is a reciprocity theorem for the generalized fingerprint problem.

Suppose that we define some functional $J(\Delta SL, \mathbf{u}, \phi)$ in terms of the solution to the fingerprint problem and wish to determine its functional derivative with respect to the direct load, ζ. To first-order, the perturbation in J can be written

$$\delta J = \int_{\partial M} \delta\Delta SL\,\zeta^\dagger \,\mathrm{d}S + \int_{\partial M} \left(\mathbf{t}^\dagger\cdot\delta\mathbf{u} + \zeta_\phi^\dagger\delta\phi\right) \mathrm{d}S, \tag{33}$$

for suitable choices of $(\zeta^\dagger, \mathbf{t}^\dagger, \zeta_\phi^\dagger)$. Eq. (32) then leads immediately to the desired relation

$$\delta J = \int_{\partial M} \Delta SL^\dagger\,\delta\zeta \,\mathrm{d}S, \tag{34}$$

with the adjoint sea level, $\Delta SL^\dagger$ obtained through the solution of the generalized fingerprint problem subject to the generalized loads $(\sigma^\dagger, \mathbf{t}^\dagger, \zeta_\phi^\dagger)$ defined within eq. (33). Here we must ask whether the

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problem for $\Delta SL^{\dagger}$ has a well-defined solution, this requiring

$$\int_{\partial M} \left(\mathbf{t}^{\dagger} - \zeta_{\phi}^{\dagger} \nabla \Phi \right) \cdot (\mathbf{a} + \mathbf{b} \times \mathbf{x}) \, \mathrm{d}S = 0, \tag{35}$$

for any constant vectors $\mathbf{a}$ and $\mathbf{b}$. Clearly, however, this condition is equivalent to the requirement that the value of J is invariant under linearized rigid body motions. Equally, the solution to the generalized fingerprint problem is only defined up to a linearized rigid body motion, but such a term makes no contribution to the resulting sea level change, $\Delta SL^{\dagger}$. The end result is that the sensitivity kernels are well-defined so long as we work with observables that are physically meaningful.

3 SENSITIVITY KERNELS IN ROTATING EARTH MODELS

Having established the basic methods in a non-rotating earth model, we show in this section how rotational feedbacks can be incorporated into the problem. Doing this requires a brief review of rotational dynamics for a deforming planet, but once this is done we can proceed very quickly to the final results.

3.1 Review of some rotational dynamics

Consider an earth model rotating steadily with angular velocity, $\boldsymbol{\Omega}$. The steady-state Euler equation takes the form

$$\boldsymbol{\Omega} \times (\mathbf{C} \cdot \boldsymbol{\Omega}) = \mathbf{0}, \tag{36}$$

where $\mathbf{C}$ is the inertia tensor. As is well-known, this equation requires that $\boldsymbol{\Omega}$ be along a principle axis of $\mathbf{C}$, and we assume that this is the axis having the largest moment of inertia, C_3. Suppose that the earth model is deformed such that its inertia tensor is perturbed to $\mathbf{C} + \mathbf{c}$. The steady-state rotation vector will be perturbed to $\boldsymbol{\Omega} + \boldsymbol{\omega}$, and from the Euler equation we obtain to, first-order accuracy,

$$\boldsymbol{\Omega} \times \mathbf{j} + C_3 \boldsymbol{\omega} \times \boldsymbol{\Omega} + \boldsymbol{\Omega} \times (\mathbf{C} \cdot \boldsymbol{\omega}) = \mathbf{0}, \tag{37}$$

where for convenience we have set $\mathbf{j} = \mathbf{c} \cdot \boldsymbol{\Omega}$. Taking the cross product with $\boldsymbol{\Omega}$ and using standard vector identities, we find that

$$(C_3 \mathbf{1} - \mathbf{C}) \cdot \boldsymbol{\omega}_{\perp} = \mathbf{j}_{\perp}, \tag{38}$$

where $\mathbf{1}$ denotes the identity tensor, and the subscript $_{\perp}$ is used to denote projections orthogonal to the reference rotation axes; note that when restricted to such vectors the matrix $C_3 \mathbf{1} - \mathbf{C}$ is non-singular. To fix the component of $\boldsymbol{\omega}$ along the rotation axes, we note that conservation of angular momentum requires

$$\| (\mathbf{C} + \mathbf{c}) \cdot (\boldsymbol{\Omega} + \boldsymbol{\omega}) \|^2 = \| \mathbf{C} \cdot \boldsymbol{\Omega} \|^2, \tag{39}$$

which to first-order yields

$$- C_3 \boldsymbol{\omega}_{\parallel} = \mathbf{j}_{\parallel}, \tag{40}$$

where the subscript $_{\parallel}$ is used to denote projections parallel to the reference rotation axes. These results can be summarized by saying that

$$\tilde{\mathbf{C}} \cdot \boldsymbol{\omega} = \mathbf{j}, \tag{41}$$

with $\tilde{\mathbf{C}}$ a symmetric and invertible matrix. Relative to a Cartesian basis aligned with the principle axes of $\mathbf{C}$, we note that

$$\tilde{\mathbf{C}} = \begin{pmatrix} C_3 - C_1 & 0 & 0 \\ 0 & C_3 - C_2 & 0 \\ 0 & 0 & -C_3 \end{pmatrix}, \tag{42}$$

where $C_1 \leq C_2 < C_3$ are the principle moments of inertia. Because $C_3 \gg C_3 - C_1 \geq C_3 - C_2$, it is common to neglect the component of $\boldsymbol{\omega}$ along the x_3-axis. This approximation will be made within our numerical calculations, but the reciprocity relations derived below hold whether or not this is done.

Associated with the equilibrium rotation vector we can define the centrifugal potential

$$\Psi = -\frac{1}{2} \| \boldsymbol{\Omega} \times \mathbf{x} \|^2, \tag{43}$$

and hence the first-order perturbation, ψ, to this potential is given by

$$\psi = -(\boldsymbol{\Omega} \times \mathbf{x}) \cdot (\boldsymbol{\omega} \times \mathbf{x}). \tag{44}$$

The negative gradient of the centrifugal potential gives the centrifugal acceleration that is needed within the equations of motion, and so we note the following relations:

$$\nabla \Psi = \boldsymbol{\Omega} \times (\boldsymbol{\Omega} \times \mathbf{x}), \quad \nabla \psi = \boldsymbol{\Omega} \times (\boldsymbol{\omega} \times \mathbf{x}) + \boldsymbol{\omega} \times (\boldsymbol{\Omega} \times \mathbf{x}). \tag{45}$$

We now need to relate an applied surface load and associated deformation of the earth model to the perturbed inertia tensor, $\mathbf{c}$. As a starting point, we note that the earth model's equilibrium inertia tensor satisfies the identity

$$\boldsymbol{\Omega} \cdot (\mathbf{C} \cdot \boldsymbol{\Omega}) = \int_M \rho \| \boldsymbol{\Omega} \times \mathbf{x} \|^2 \, \mathrm{d}V, \tag{46}$$

with ρ the equilibrium density, and where $\boldsymbol{\Omega}$ here is an arbitrary vector. When subject to a surface load, σ, the first-order perturbation in this quantity can be written

$$\boldsymbol{\Omega} \cdot (\mathbf{c} \cdot \boldsymbol{\Omega}) = - \int_M 2 \rho \mathbf{u} \cdot [\boldsymbol{\Omega} \times (\boldsymbol{\Omega} \times \mathbf{x})] \, \mathrm{d}V + \int_{\partial M} \sigma \| \boldsymbol{\Omega} \times \mathbf{x} \|^2 \, \mathrm{d}S, \tag{47}$$

where standard vector identities have been applied. Here, the first term is due to internal deformation and the second is the direct contribution of the load. Changing $\boldsymbol{\Omega}$ within this identity to $\boldsymbol{\Omega} + \boldsymbol{\omega}'$ and expanding to first-order in $\boldsymbol{\omega}'$ we obtain

$$\boldsymbol{\omega}' \cdot \mathbf{j} = - \int_M \rho \mathbf{u} \cdot [\boldsymbol{\omega}' \times (\boldsymbol{\Omega} \times \mathbf{x}) + \boldsymbol{\Omega} \times (\boldsymbol{\omega}' \times \mathbf{x})] \, \mathrm{d}V + \int_{\partial M} \sigma (\boldsymbol{\Omega} \times \mathbf{x}) \cdot (\boldsymbol{\omega}' \times \mathbf{x}) \, \mathrm{d}S, \tag{48}$$

where we recall that $\mathbf{j} = \mathbf{c} \cdot \boldsymbol{\Omega}$. Comparing the two integrands with the expression for ψ and $\nabla \psi$ obtained above, we arrive at an identity that will be useful later

$$\boldsymbol{\omega}' \cdot \mathbf{j} + \int_M \rho \mathbf{u} \cdot \nabla \psi' \, \mathrm{d}V + \int_{\partial M} \sigma \psi' \, \mathrm{d}S = 0. \tag{49}$$

Within this equality it should be emphasized that ψ' is the centrifugal potential perturbation associated with $\boldsymbol{\omega}'$, while $\mathbf{j}$ is the inertia perturbation associated with the applied load, σ, and the associated displacement, $\mathbf{u}$.

The derivations within this section have assumed that the earth model is entirely solid. As shown by Dahlen (1974), in considering the static deformation of an earth model with a fluid (outer) core, the linearized Lagrangian displacement vector is not well-defined in fluid regions. Instead, Dahlen showed that an Eulerian formulation within fluid regions could be applied, with all relevant variables then expressed in terms of the perturbed gravitational potential. The necessary extensions of our arguments to earth models containing fluid regions are simple but will not be given because the form of the reciprocity theorems is unchanged.

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It is also worth commenting that, when calculations are performed in a spherically symmetric earth model, the reference inertia tensor, $\mathbf{C}$, is taken to be that observed for the Earth (*cf.* Mitrovica *et al.* 2005; Mitrovica & Wahr 2011). Here, there is an assumption that the deformation of the Earth due to the applied load can be sufficiently well-approximated by the response of a spherical earth model. Again, the reciprocity theorems hold whether or not this approximation is made.

3.2 Reciprocity theorems including rotational feedbacks

We start with the equations for a static loading problem in a rotating earth model in weak form,

$$\mathcal{A}(\mathbf{u}, \phi \,|\, \mathbf{u}', \phi') + \int_{\partial M} (\mathbf{u}' \cdot \nabla\Phi + \phi')\,\sigma \,\mathrm{d}S + \int_M \rho\mathbf{u}' \cdot \nabla\psi \,\mathrm{d}V + (\tilde{\mathbf{C}} \cdot \boldsymbol{\omega} - \mathbf{j}) \cdot \boldsymbol{\omega}' = 0. \tag{50}$$

Here, ψ is the centrifugal potential associated with the perturbed rotation vector, $\boldsymbol{\omega}$, and we see that this term is associated with a tidal force applied to the earth model (*cf.* Bagheri *et al.* 2019, Appendix B). Note also that a new test function, $\boldsymbol{\omega}'$, has been added to enforce the relation between $\boldsymbol{\omega}$ and the inertia tensor perturbation $\mathbf{j}$ in eq. (41), the latter term depending implicitly on $\mathbf{u}$ and the applied load, σ. Following a now familiar procedure, we let $(\mathbf{u}^\dagger, \phi^\dagger, \boldsymbol{\omega}^\dagger)$ be the solution of the problem subject to a load $\sigma^\dagger$, and use this to obtain the identity

$$\int_{\partial M} \left(\mathbf{u}^\dagger \cdot \nabla\Phi + \phi^\dagger\right) \sigma \,\mathrm{d}S + \int_M \rho\mathbf{u}^\dagger \cdot \nabla\psi \,\mathrm{d}V + \mathbf{j}^\dagger \cdot \boldsymbol{\omega} = \int_{\partial M} (\mathbf{u} \cdot \nabla\Phi + \phi)\,\sigma^\dagger \,\mathrm{d}S + \int_M \rho\mathbf{u} \cdot \nabla\psi^\dagger \,\mathrm{d}V + \mathbf{j} \cdot \boldsymbol{\omega}^\dagger, \tag{51}$$

where we note that terms involving the matrix $\tilde{\mathbf{C}}$ have vanished due to its symmetry. Using eq. (49) twice, this simplifies to

$$\int_{\partial M} \left(\mathbf{u}^\dagger \cdot \nabla\Phi + \phi^\dagger + \psi^\dagger\right) \sigma \,\mathrm{d}S = \int_{\partial M} (\mathbf{u} \cdot \nabla\Phi + \phi + \psi)\,\sigma^\dagger \,\mathrm{d}S. \tag{52}$$

This is a new result which extends the reciprocity theorem of Tromp & Mitrovica (1999) to incorporate rotational feedbacks. It is worth commenting that analogous reciprocity theorems can be established for elastodynamics (e.g. Dahlen & Tromp 1998), but in that case it is necessary to consider equations of motion in which the sign of the rotation vector, $\boldsymbol{\Omega}$, has been changed. Such a sign change arises due to the Coriolis term within the equations of motion being anti-self-adjoint, but this term is absent within the static equations we consider.

To express eq. (52) in terms of sea level change, we need merely note that, once rotational feedbacks are present, the relation between sea level change, ΔSL, and surface deformation is generalized to

$$\Delta SL = -\frac{1}{g}(\mathbf{u} \cdot \nabla\Phi + \phi + \psi) + \frac{\Phi_g}{g}, \tag{53}$$

which follows due to the sea surface now lying on an equipotential of the combined gravitational and centrifugal potentials. Given this relation, our earlier argument proceeds identically to give

$$\int_{\partial M} \Delta SL^\dagger\, \zeta \,\mathrm{d}S = \int_{\partial M} \Delta SL\, \zeta^\dagger \,\mathrm{d}S, \tag{54}$$

which takes the same form as in the non-rotating case in eq. (17). From this identity we can readily show the symmetry of the Green's function and obtain sensitivity kernels for any functional of the sea level change. These results are new, extending the work of Crawford *et al.* (2018) to include rotational feedbacks in the case of elastic fingerprint problems. Extensions of the general viscoelastic adjoint theory to incorporate rotational feedbacks can be carried out along similar lines, but the results will be presented elsewhere.

Finally, if we wish to consider observables other than sea level, it is necessary to consider the generalized loading problem

$$\mathcal{A}(\mathbf{u}, \phi \,|\, \mathbf{u}', \phi') + \int_{\partial M} (\mathbf{u}' \cdot \nabla\Phi + \phi')\,\sigma \,\mathrm{d}S + \int_{\partial M} (\mathbf{t} \cdot \mathbf{u}' + \zeta_\phi \phi') \,\mathrm{d}S + \int_M \rho\mathbf{u}' \cdot \nabla\psi \,\mathrm{d}V + (\tilde{\mathbf{C}} \cdot \boldsymbol{\omega} - \mathbf{j} + \mathbf{k}) \cdot \boldsymbol{\omega}' = 0, \tag{55}$$

where an additional force term, $\mathbf{k}$, has been included to facilitate observations depending directly on $\boldsymbol{\omega}$. Here, we must account for the contribution of the direct gravitational load, ζ_ϕ, to the perturbed inertia tensor, this leading to eq. (49) being generalized to

$$\boldsymbol{\omega}' \cdot \mathbf{j} + \int_M \rho\mathbf{u} \cdot \nabla\psi' \,\mathrm{d}V + \int_{\partial M} (\sigma + \zeta_\phi)\psi' \,\mathrm{d}S = 0. \tag{56}$$

Following the now standard argument, we arrive at the reciprocity theorem

$$\int_{\partial M} \Delta SL^\dagger\, \zeta \,\mathrm{d}S - \frac{1}{g}\int_{\partial M} [\mathbf{t} \cdot \mathbf{u}^\dagger + \zeta_\phi(\phi^\dagger + \psi^\dagger)] \,\mathrm{d}S - \frac{1}{g}\mathbf{k} \cdot \boldsymbol{\omega}^\dagger = \int_{\partial M} \Delta SL\, \zeta^\dagger \,\mathrm{d}S - \frac{1}{g}\int_{\partial M} [\mathbf{t}^\dagger \cdot \mathbf{u} + \zeta_\phi^\dagger(\phi + \psi)] \,\mathrm{d}S - \frac{1}{g}\mathbf{k}^\dagger \cdot \boldsymbol{\omega}. \tag{57}$$

To remove the explicit dependence on the centrifugal potentials, we note from eq. (44) that $\psi = -[\mathbf{x} \times (\boldsymbol{\Omega} \times \mathbf{x})] \cdot \boldsymbol{\omega}$ and hence

$$\begin{aligned}\int_{\partial M} \Delta SL^\dagger\, \zeta \,\mathrm{d}S &- \frac{1}{g}\int_{\partial M} (\mathbf{t} \cdot \mathbf{u}^\dagger + \zeta_\phi \phi^\dagger) \,\mathrm{d}S \\ &- \frac{1}{g}\left[\mathbf{k} - \int_{\partial M} \zeta_\phi\, \mathbf{x} \times (\boldsymbol{\Omega} \times \mathbf{x}) \,\mathrm{d}S\right] \cdot \boldsymbol{\omega}^\dagger \\ &= \int_{\partial M} \Delta SL\, \zeta^\dagger \,\mathrm{d}S - \frac{1}{g}\int_{\partial M} (\mathbf{t}^\dagger \cdot \mathbf{u} + \zeta_\phi^\dagger \phi) \,\mathrm{d}S \\ &\quad - \frac{1}{g}\left[\mathbf{k}^\dagger - \int_{\partial M} \zeta_\phi^\dagger\, \mathbf{x} \times (\boldsymbol{\Omega} \times \mathbf{x}) \,\mathrm{d}S\right] \cdot \boldsymbol{\omega},\end{aligned} \tag{58}$$

which is an alternative form that can sometimes be useful.

4 NUMERICAL VERIFICATION AND INITIAL APPLICATIONS

Within this final section, we describe a range of numerical calculations that (i) provide a check on the correctness our theoretical results and (ii) illustrate the potential of these methods within future practical applications. All these calculations are done using a spherically symmetric earth model, but we note that the general theory is applicable within laterally varying elastic earth models. The use of spherically symmetric earth models is common within fingerprint calculations and it has been shown by Mitrovica *et al.* (2011) that lateral variations in elastic structure have only a minor effect in this context.

4.1 Summary of the numerical method

To solve the elastic fingerprint problem, we apply the standard pseudo-spectral method (Mitrovica & Peltier 1991) with rotational feedbacks incorporated following Milne & Mitrovica (1996, 1998). Fast spherical harmonic transformations are performed using the SHTOOLS library of Wieczorek & Meschede (2018). Within all

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calculations, a truncation degree of 512 is used, but it has been verified that truncation at a higher degree does not appreciably change any of the results.

To account for deformation of the solid earth, we use loading and tidal Love numbers (Love 1911). For example, for a surface load, σ, that has spherical harmonic decomposition

$$\sigma = \sum_{lm} \sigma_{lm}\, Y_{lm}, \tag{59}$$

the resulting surface vertical displacement, u, and gravitational potential, ϕ, have coefficients

$$u_{lm} = h_l\, \sigma_{lm}, \quad \phi_{lm} = k_l\, \sigma_{lm}, \tag{60}$$

with h_l and k_l the loading Love numbers (as defined in this work). Here Y_{lm} denotes an orthonormalized real spherical harmonic of degree l and order m defined following Wieczorek & Meschede (2018) (note that the definition used for these functions does not include the Condon–Shortley phase).

The generalized fingerprint problem involves additional like-load terms, $\mathbf{t}$ and ζ_ϕ, introduced within eq. (29). Within the following examples, we limit attention to functionals whose dependence on the surface displacement is only through the vertical component. It is, therefore, sufficient to assume

$$\mathbf{t} = \zeta_u \nabla \Phi, \tag{61}$$

for a scalar-valued function ζ_u. We can then proceed by introducing suitably generalized loading Love numbers. If ζ^u_{lm} and ζ^ϕ_{lm} are spherical harmonic expansion coefficients of ζ_u and ζ_ϕ, the associated coefficients for u and ϕ can be written

$$u_{lm} = h_l\, \sigma_{lm} + h^u_l\, \zeta^u_{lm} + h^\phi_l\, \zeta^\phi_{lm}, \quad \phi_{lm} = k_l\, \sigma_{lm} + k^u_l\, \zeta^u_{lm} + k^\phi_l\, \zeta^\phi_{lm}, \tag{62}$$

where h^u_l, h^ϕ_l, k^u_l and k^ϕ_l are the generalized loading Love numbers. Calculation of these Love numbers proceeds in the same manner as usual, with the only difference being the boundary conditions applied when solving the elastostatic equations. From these definitions we note the relations

$$h_l = h^u_l + h^\phi_l, \quad k_l = k^u_l + k^\phi_l, \tag{63}$$

while, by applying eq. (31), we also have

$$g\, h^\phi_l = k^u_l, \tag{64}$$

which provides a useful check within numerical calculations. Values for these generalized loading Love numbers along with the usual tidal Love numbers have been calculated for PREM (Dziewonski & Anderson 1981) up to degree 4096 using the radial spectral element method of Al-Attar & Tromp (2014). Translational non-uniqueness within the degree-one problem is removed by requiring that the combined centre of mass of the earth model and surface loads is fixed. The remaining rotational degrees of freedom do not affect the observables we consider, and hence they can be ignored.

4.2 Generating a synthetic data set

Within Fig. 1 we summarize the synthetic data set used in later examples. For the purposes of this paper, the specific change in ice thickness that has been chosen is arbitrary, but for definiteness it was constructed as follows. Within the northern and southern hemispheres the changes in ice thickness were separately set to be constant fractions of the present-day ice thickness within ICE6G (Argus *et al.* 2012; Peltier *et al.* 2015). These two contributions were then linearly combined such that 20 per cent of the ice mass loss was sourced from the northern hemisphere, while the overall magnitude was fixed by requiring the global mean sea level change equal 5 mm. Using this change in ice thickness, ΔI, as an input, the sea level equation was solved to obtain the resulting sea level change, ΔSL, vertical displacement, u, gravitational potential perturbation, ϕ, rotation vector, $\boldsymbol{\omega}$ and centrifugal potential perturbation, ψ, with Fig. 1 also showing the first three of these outputs.

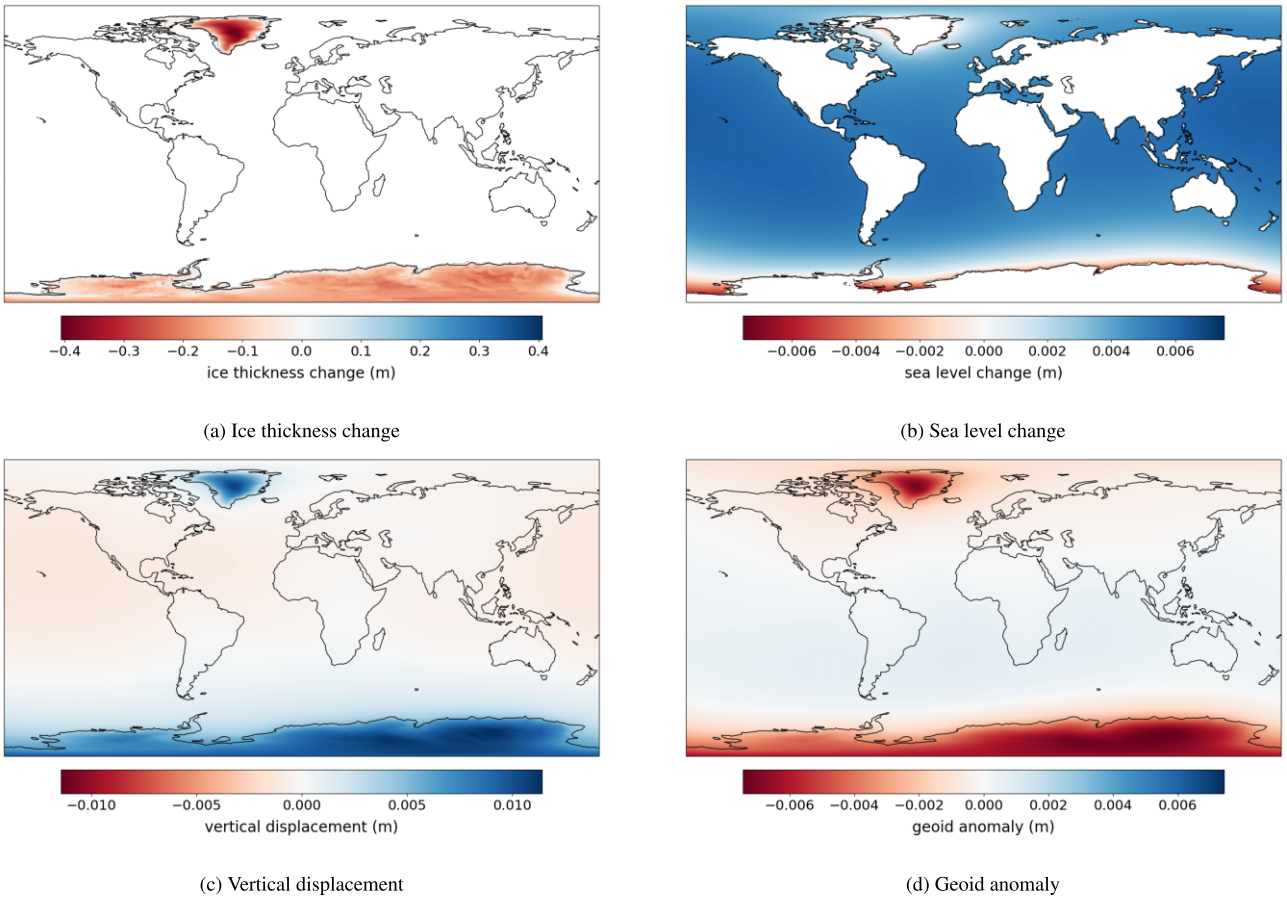
4.3 Sensitivity kernels for point measurements of sea level

As noted previously, the sensitivity kernel for a point measurement of sea level at a location $\mathbf{x}_0$ can be obtained taking the adjoint load, $\zeta^\dagger$, to be a delta function at the observation point. Within Fig. 2 we show the results of such a calculation for Boston, Massachusetts. Within subfigure (a), the sensitivity kernel, $\Delta SL^\dagger$, for this measurement with respect to the direct load, ζ, is plotted. As might be expected, this sensitivity is greatest at the observation point and falls in magnitude rapidly away from it. The value of the kernel at the observation point is positive. This means that an increase of the direct load in the vicinity of Boston would be associated with a local rise in sea level. Such behaviour is, of course, consistent with the known physics of sea level change.

To check quantitatively whether this kernel is correct, we can numerically evaluate and compare either side of eq. (54) using the load and sea level shown in Fig. 1. For the left-hand side of the identity, we evaluate $\int_{\partial M} \Delta SL^\dagger\, \zeta \, \mathrm{d}S$ numerically using the solution, $\Delta SL^\dagger$, of the adjoint problem along with the direct load, ζ, applied within the forward problem. The right-hand side of the identity is given by $\int_{\partial M} \Delta SL\, \zeta^\dagger \, \mathrm{d}S$, and this can be determined by numerical integration or simply through evaluation of ΔSL at the observation point. The relative difference between the values obtained was of order 10^{-7} which is about the accuracy of the numerical calculations (no formal error analysis of the numerical methods has been undertaken). This test has been repeated using a range of different direct loads, while similar comparisons have been made for each the kernels corresponding to each of the observables discussed below.

The sensitivity kernel of sea level with respect to ice thickness is given by $\rho_\mathrm{i}\,(1 - C)\,\Delta SL^\dagger$ which is plotted in Fig. 2(b) for the Boston measurement. Here it is worth noting that this kernel can be non-zero anywhere on land. It makes sense, however, in most applications to multiply this kernel by a function that equals one in glaciated regions and by zero elsewhere. This has been done in Fig. 1(c), and now we are able to see more interesting spatial variability. This kernel does, however, contain some high-wavenumber oscillations, with this feature being more pronounced when lower truncation degrees are used within the calculations. The reason is that a delta function cannot be accurately represented using truncated spherical harmonic expansions (nor in any finite-dimensional basis). For some applications these oscillations have no practical effect and can be ignored. However, a simple and pragmatic way to remove the oscillations is to replace the point load by an appropriate smooth averaging function (*cf.* Al-Attar & Tromp 2014, Appendix E). The resulting kernel is shown in Fig. 1(d) obtained using an averaging function with a characteristic length-scale of one degree (in more detail, the Green's function for the heat equation on a sphere was used, with the time chosen such that diffusion over the desired length-scale has occurred). The result of this latter calculation can be compared with fig. 1 in Mitrovica *et al.* (2018) which shows a sensitivity kernel for sea level at Boston (along with several other locations) to variations in ice thickness over Greenland. Within that paper, a finite-difference approach was used to obtain

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(a) Ice thickness change

(b) Sea level change

(c) Vertical displacement

(d) Geoid anomaly

Figure 1. A summary of the input and results of the forward fingerprint calculation used within most of the later numerical examples. In (a) the chosen ice thickness change is shown. In (b) we plot the resulting sea level change obtained through solution of a fingerprint problem. Note that while values of the sea level change are only plotted here within the oceans, this field is defined globally. As part of that calculation, we also determined the vertical displacement, $\hat{\mathbf{n}} \cdot \mathbf{u}$, which is shown in (c), and the gravitational potential perturbation, ϕ, which is shown in (d) as the associated geoid anomaly, $-\phi/g$. Values for the perturbed rotation vector, $\boldsymbol{\omega}$, and associated centrifugal potential, ψ, were also determined but have not been plotted.

these sensitivity kernels, and this required many forward calculations. Using our present methods such kernels can be determined for any desired location through a single fingerprint calculation.

4.4 Sensitivity kernels for vertical displacement measurements

Consider a measurement of vertical displacement at a surface location $\mathbf{x}_0$, made using the Global Positioning System (GPS) (e.g. Khan *et al.* 2010). We can determine the sensitivity kernel for such a measurement with respect to the direct load by solving the generalized fingerprint problem subject to the following adjoint loads:

$$\zeta^\dagger = 0, \quad \zeta_u^\dagger(\mathbf{x}) = -\delta(\mathbf{x}, \mathbf{x}_0), \quad \zeta_\phi^\dagger = 0, \quad \mathbf{k}^\dagger = \mathbf{0}. \tag{65}$$

The result of two such calculations is shown in Fig. 3 for SENU station in Greenland and DUPT in Antarctica. Here, as before, we have used smoothed point loads to remove non-physical oscillations from the kernel. It should be emphasized that in these examples, the displacements are computed relative to the combined centre of mass of earth model and surface loads. A translation of these results to any other reference system (e.g. the centre of mass of the solid Earth) is straightforward. The aim of these examples is merely to show that the calculation of such sensitivity kernels can be readily done, with the practical application of these ideas to be discussed in future work.

4.5 Sensitivity kernels for gravitational potential coefficients

We consider the measurement of a spherical harmonic coefficient of the gravitational potential at degree $l \geq 2$ as might be obtained using GRACE or GRACE-FO (e.g. Tapley *et al.* 2004; Landerer *et al.* 2020). For definiteness, we use real, fully normalized spherical harmonics, denoting the (l, m)th function by Y_{lm}. In determining the adjoint loads for this application, we use eq. (58) to select a value for $\mathbf{k}^\dagger$ such that the dependence on $\boldsymbol{\omega}$ is removed from the right-hand side, with the result:

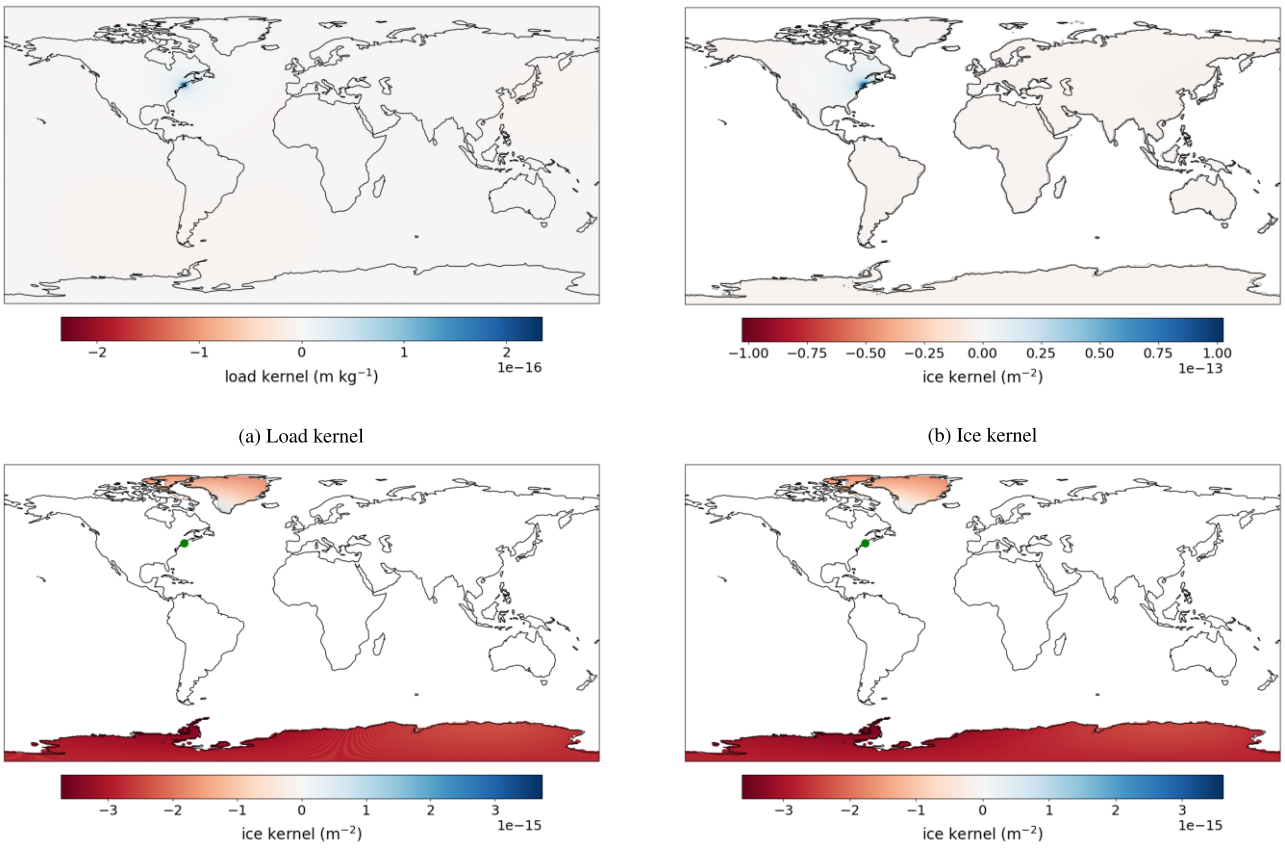
$$\zeta^\dagger = 0, \quad \zeta_u^\dagger(\mathbf{x}) = 0, \quad \zeta_\phi^\dagger = -g\, Y_{lm}, \quad \mathbf{k}^\dagger = \int_{\partial M} \zeta_\phi^\dagger\, \mathbf{x} \times (\boldsymbol{\Omega} \times \mathbf{x})\, \mathrm{d}S. \tag{66}$$

Example sensitivity kernels for $l = 4$ and $m = 2$ are shown in Fig. 4. The kernel with respect to ζ is plotted globally in (a), whereas that for ice thickness is projected onto glaciated regions in (b).

The kernels just shown reflect contributions to the potential coefficient associated with both the direct load and the gravitationally self-consistent water load it must induce. They can be usefully contrasted with kernels that account only for the direct load. In the latter case we can simply write

$$\phi_{lm} = k_l\, \zeta_{lm}, \tag{67}$$

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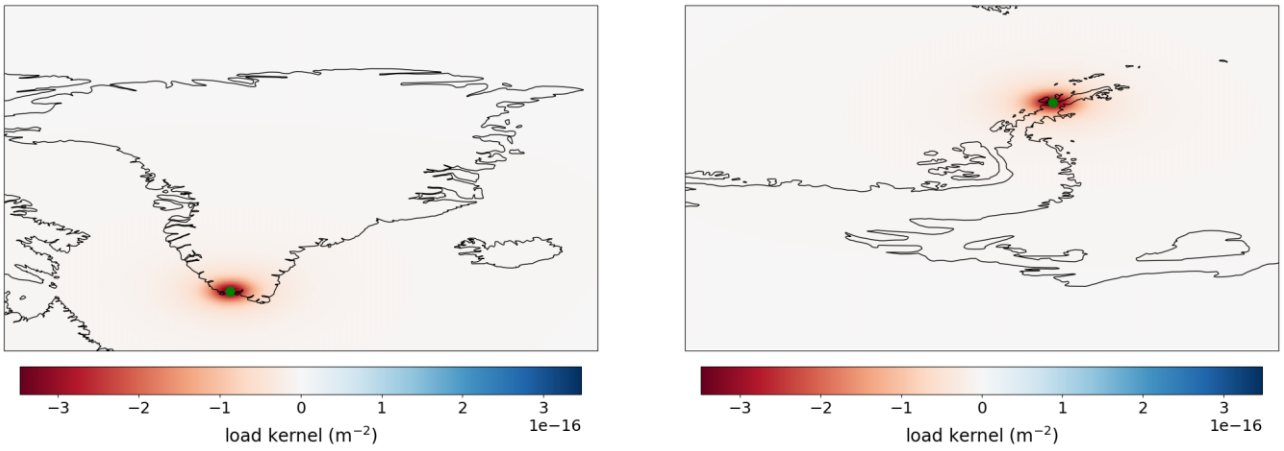
(a) Load kernel

(b) Ice kernel

(c) Ice kernel within glaciated regions

(d) Smoothed ice kernel within glaciated regions

Figure 2. Sensitivity kernels for a point measurement of sea level change in Boston, Massachusetts. In (a) we show the kernel with respect to the direct load, ζ, this being equal to the adjoint sea level, $\Delta SL^{\dagger}$ obtained by solving the fingerprint problem for a point load at Boston. In (b), we show the associated sensitivity kernel with respect to ice thickness which takes the form $\rho_{\mathrm{i}}(1-C)\Delta SL^{\dagger}$. Note that this kernel is non-zero in all locations where grounded ice *could* be present. For most applications, it makes sense to project this kernel onto currently glaciated regions, and this is done in (c). Here we can now see more structure in the kernel, but also a slight ringing associated with the use of truncated spherical harmonic expansions. A pragmatic way to avoid this issue is to replace the point load with a smooth average over an appropriate length scale. The resulting sensitivity kernel is shown in (d) with the measurement corresponding an average of the sea level within about one degree of Boston. Note that in (c) and (d) the observation point is shown as a green dot.

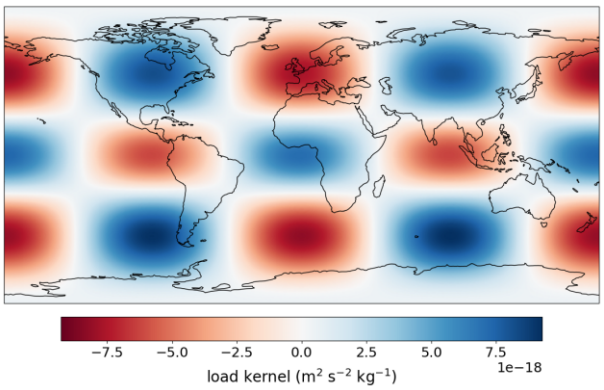


(a) Load kernel for SENU

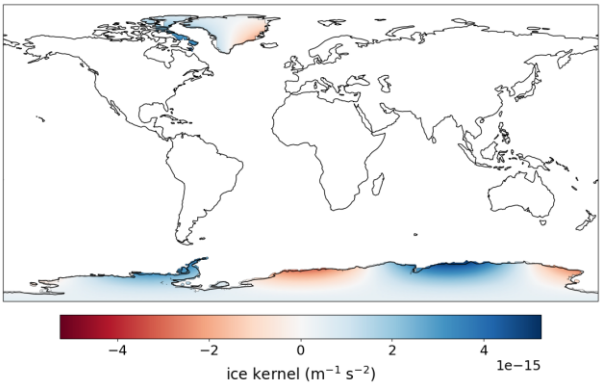
(b) Load kernel for DUPT

Figure 3. The sensitivity kernel of vertical displacement at the GPS stations (a) SENU in Greenland and (b) DUPT in Antarctica with respect to direct load. As before, the point load occurring within the generalized fingerprint problem has been smoothed over a distance of one degree to remove high-wavenumber oscillations from the kernel. In each plot, the observation point is indicated by a green dot.

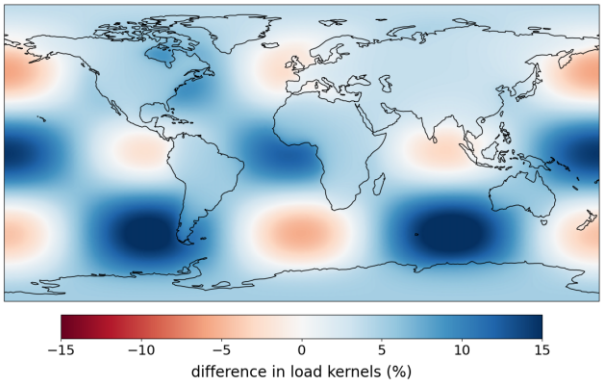
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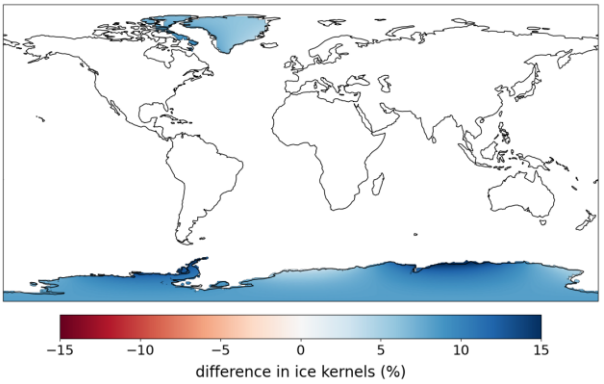
(a) Load kernel



(b) Ice kernel



(c) Difference with load kernel that neglects induced water loading



(d) Difference with ice kernel that neglects induced water loading

Figure 4. In (a) we show the sensitivity kernel for the spherical harmonic coefficient, $\phi_{4,2}$, of the gravitational potential perturbation to the direct load, ζ. The kernel for the same measurement but with respect to ice thickness over glaciated regions is then plotted in (b). Within the calculation of these sensitivity kernels, the effect of both the direct load and a gravitationally self-consistent water load have been taken into account. Their form is compared in (c) and (d) to simpler kernels that account only for the direct load. In (c) we plot the difference between the kernel in (a) and this simpler kernel defined with respect to ζ. Values are shown as a percentage defined relative to the maximum absolute value of the kernel in (a). In (d) we show the analogous difference for the kernel with respect to ice thickness over glaciated regions.

with k_l the loading Love number for ϕ at degree l. Given this relation, we then have

$$\phi_{lm} = \frac{k_l}{b^2} \int_{\partial M} \zeta \, Y_{lm} \, \mathrm{d}S, \tag{68}$$

with b the radius of the earth model, and hence we conclude that the relevant kernel is $\frac{k_l}{b^2} Y_{lm}$. In Fig. 4(c), we show the relative difference between the sensitivity kernel with respect to the direct load in (a) and the simpler kernel just defined. As might be expected, the relative differences are largest within the oceans, but even on land they can be of the order of several percent. Finally, in (d) we plot the corresponding difference between the ice kernels over glaciated regions, observing that its magnitude can in places be upwards of ten percent. Within Fig. 5 we show the same thing for a higher-degree coefficient of the gravitational potential, specifically $l = 20$ and $m = 5$. Here, we see smaller differences due to the neglect of induced water loads, but again the differences are most significant within the oceans. These results are consistent with the work of Sterenborg *et al.* (2013) who showed that methods for analysing GRACE or GRACE-FO data that do not account for induced water loads suffer from systematic biases. Within these examples we see, in particular, that the importance of the induced water load is scale-dependent, decreasing as the dominant wavenumber of the direct load increases.

Building on the previous example, we now consider the use of GRACE coefficients to estimate surface mass changes within ice sheets (e.g. Velicogna & Wahr 2006) following the widely used method of Wahr *et al.* (1998) and Swenson & Wahr (2002). In the notations of this paper, the approach starts with relation

$$\phi_{lm} = k_l \, \sigma_{lm}, \tag{69}$$

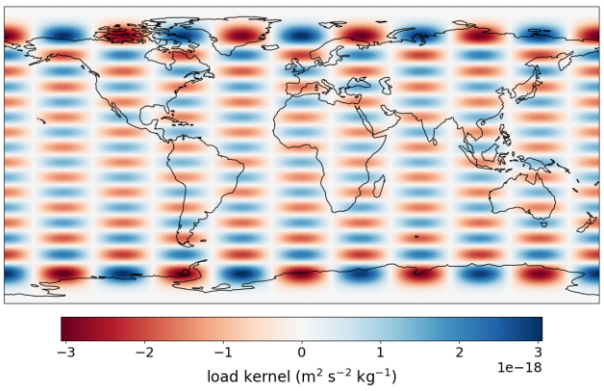
between the spherical harmonic coefficients of the surface load and those of the resulting gravitational potential perturbation. Here, by use of the appropriate loading Love number, we are accounting for the direct gravitational effect of the surface mass along with the contribution to ϕ associated with deformation of the earth model (*cf.* Wahr *et al.* 1998, eq. 12). We assume that the potential coefficients at degrees zero and one vanish, the former due to conservation of mass and latter because the measurements are made relative to the centre of mass frame. For $l \geq 2$, the Love numbers are non-zero and so we can write

$$\sigma_{lm} = k_l^{-1} \, \phi_{lm}, \tag{70}$$

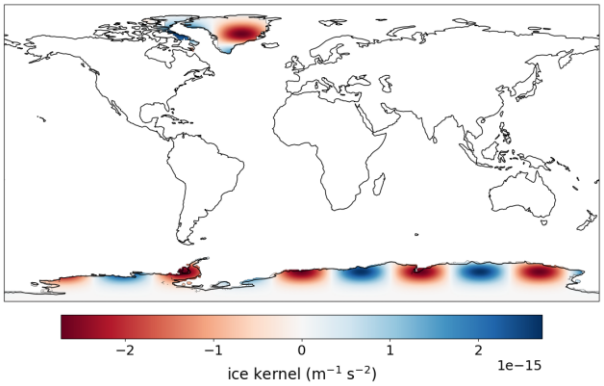
and hence reconstruct the load through

$$\sigma \approx \sum_{lm} k_l^{-1} \, \phi_{lm} \, Y_{lm}, \tag{71}$$

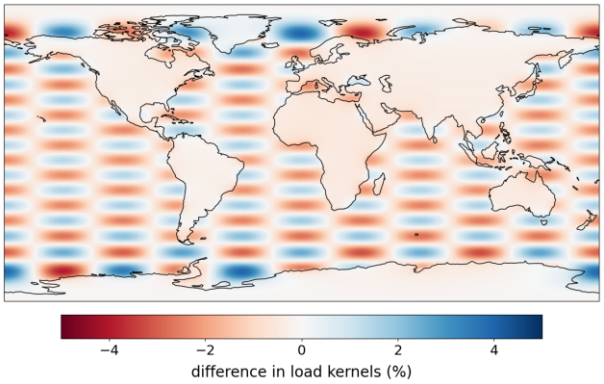
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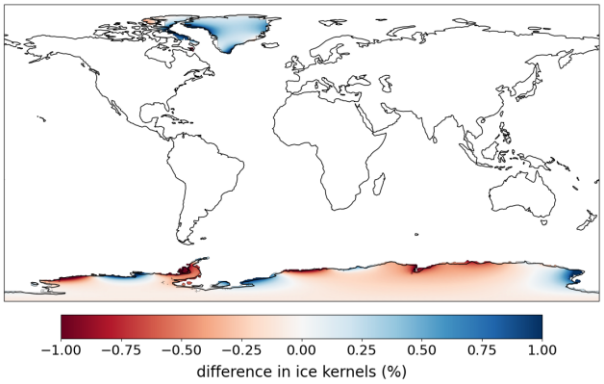
(a) Load kernel



(b) Ice kernel



(c) Difference with load kernel that neglects induced water loading



(d) Difference with ice kernel that neglects induced water loading

Figure 5. As in Fig. 4, but showing results for the $l = 20$, $m = 5$ coefficient of the gravitational potential.

where an approximately equals sign is included because terms for $l \leq 1$ are excluded. Errors in GRACE observations are known to increase quite rapidly as a function of degree (e.g. Swenson & Wahr 2002, Fig. 1), and so the summation in eq. (71) must be truncated at some finite-degree, L, with the the choice $L = 100$ being common. Using this approach one can obtain point-estimates of surface loads from GRACE data. In fact, it is often more useful to estimate average loads over a specific region of interest. This is done by introducing an averaging function, w, and then using eq. (71) to obtain

$$\int_{\partial M} w\,\sigma\,\mathrm{d}S \approx \sum_{lm} b^2 k_l^{-1}\,\phi_{lm}\,w_{lm}, \tag{72}$$

with w_{lm} the expansion coefficients of the averaging function; in what follows, we call the right-hand side of this expression the 'GRACE estimate' of the load. The design of optimal averaging functions for a region has been discussed by Swenson & Wahr (2002), but in our numerical examples we only use the simple Gaussian functions of Wahr *et al.* (1998).

Within eq. (72), the left-hand side is the quantity that we wish to estimate, namely the average of the load over a chosen region, while on the right is a something that can be computed from GRACE data. As we have noted, this equality is approximate because of the absence of information at degrees $l \leq 1$, the truncation of the summation at a finite-value, and due to the propagation of measurement errors. There is, however, another potential source of uncertainty within this method. The derivation of eq. (72) started from the relation in eq. (69) between spherical harmonic components of the load and the gravitational potential, but this does not account for the fact that any load on the Earth's surface must induce a corresponding water load. To investigate the significance of this issue, it will be useful to first rewrite the right-hand side of eq. (72)

$$\sum_{lm} b^2 k_l^{-1}\,\phi_{lm}\,w_{lm} = \int_{\partial M} \tilde{w}\,\phi\,\mathrm{d}S, \tag{73}$$

where we have defined

$$\tilde{w} = \sum_{lm} k_l^{-1}\,w_{lm}\,Y_{lm}, \tag{74}$$

with, again, the summation leaving implicit the inclusion of only degrees $2 \leq l \leq L$. From the reciprocity theorem in eq. (58), we can write

$$\int_{\partial M} \tilde{w}\,\phi\,\mathrm{d}S = \int_{\partial M} \Delta SL^{\dagger}\,\zeta\,\mathrm{d}S, \tag{75}$$

where $\Delta SL^{\dagger}$ is obtained by solving the generalized fingerprint problem subject to the adjoint loads

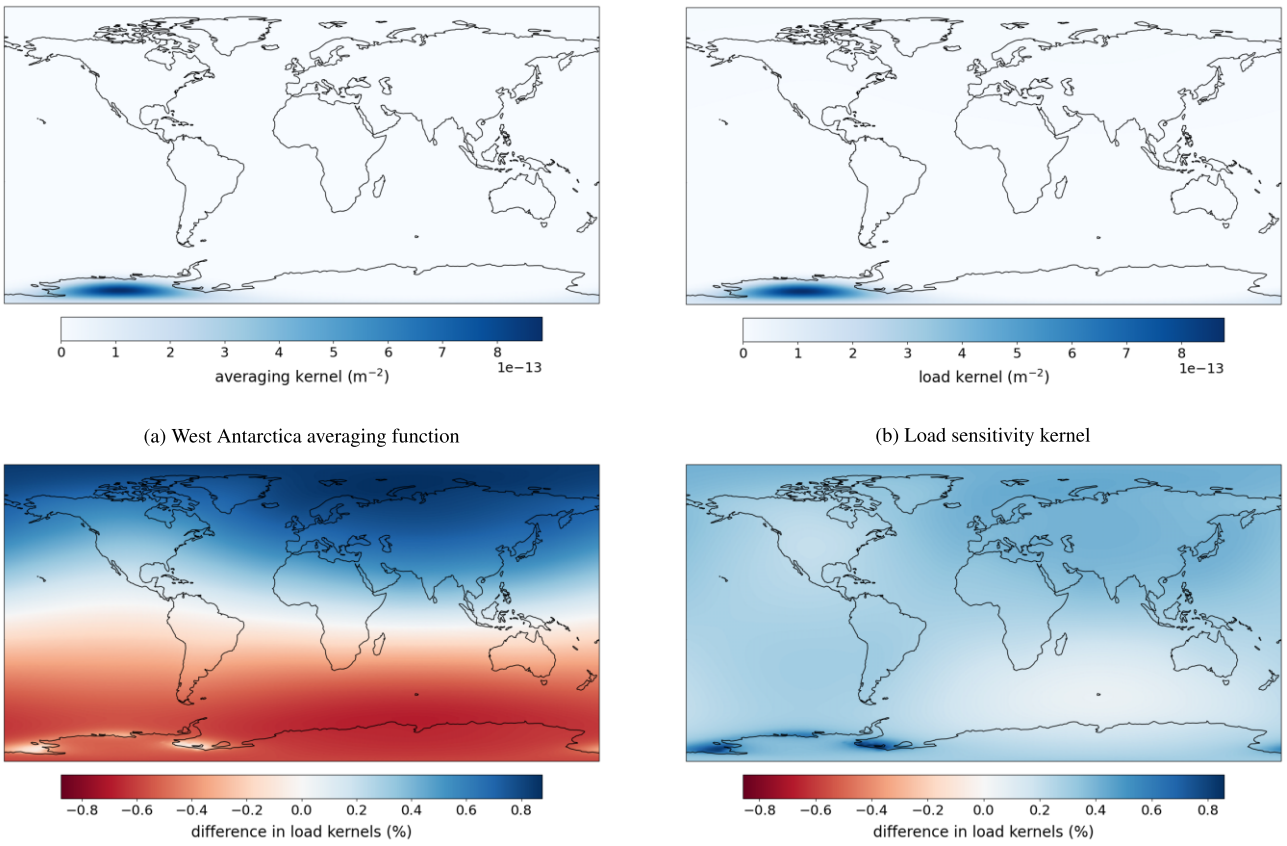
$$\zeta^{\dagger} = 0, \quad \zeta_u^{\dagger}(\mathbf{x}) = 0, \quad \zeta_{\phi}^{\dagger} = -g\,\tilde{w}, \quad \mathbf{k}^{\dagger} = \int_{\partial M} \zeta_{\phi}^{\dagger}\,\mathbf{x} \times (\boldsymbol{\Omega} \times \mathbf{x})\,\mathrm{d}S. \tag{76}$$

Using these results, we can rewrite eq. (72) in the form

$$\int_{\partial M} w\,\sigma\,\mathrm{d}S \approx \int_{\partial M} \tilde{w}\,\phi\,\mathrm{d}S = \int_{\partial M} \Delta SL^{\dagger}\,\zeta\,\mathrm{d}S. \tag{77}$$

It is notable that within this expression we have both the total load, σ, and the direct load, ζ. To simplify the discussion, we shall assume

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(a) West Antarctica averaging function

(b) Load sensitivity kernel

(c) Difference between kernel and averaging function

(d) Difference between kernel and averaging function with $l \leq 1$ removed

Figure 6. In (a) we show a Gaussian load averaging function targeted at West Antarctica with a half-width of 500 km. In (b) we plot the resulting sensitivity kernel with respect to the direct load. The relative difference between the kernel and the averaging function is shown in (c) with the values normalized by the maximum value of the latter field. In (d) we again see the relative difference between the load kernel and the averaging function, but with the calculations repeated with degrees $l \leq 1$ removed from the averaging function.

that the load average is defined over a region on land, and hence we can replace σ with ζ on the left-hand side to obtain

$$\int_{\partial M} w\,\zeta\,\mathrm{d}S \approx \int_{\partial M} \tilde{w}\,\phi\,\mathrm{d}S = \int_{\partial M} \Delta SL^{\dagger}\,\zeta\,\mathrm{d}S. \tag{78}$$

Here, in the middle, we have the term that is computed from the GRACE data. The left-hand term is the load average we wish to estimate, while on the right is the function of the direct load we are actually calculating. It follows that the accuracy of the GRACE estimate is determined by the extent to which the load kernel, $\Delta SL^{\dagger}$, resembles the averaging function, w. If, for example, the equality $w = \Delta SL^{\dagger}$ were to hold, then we would have

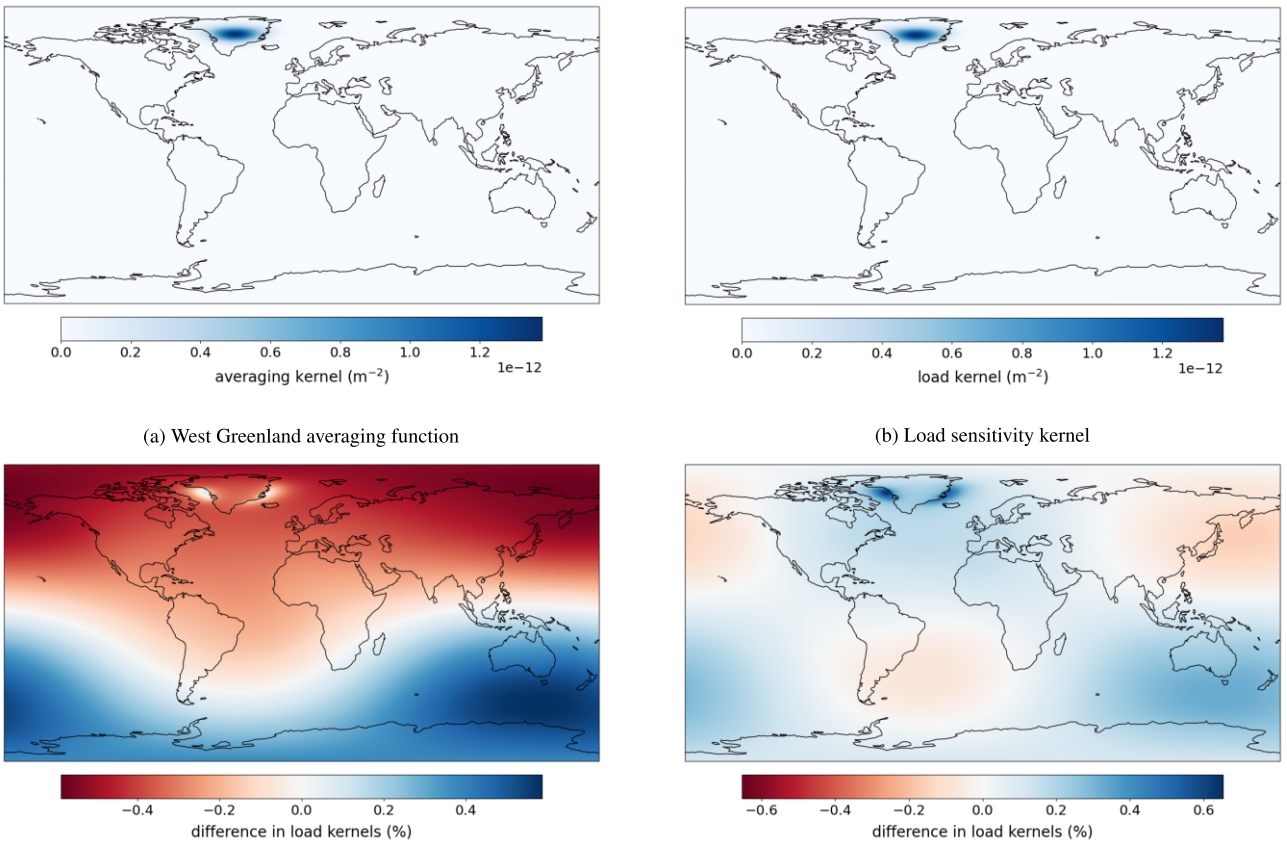
$$\int_{\partial M} w\,\zeta\,\mathrm{d}S = \int_{\partial M} \tilde{w}\,\phi\,\mathrm{d}S, \tag{79}$$

meaning the GRACE estimate is exact in the absence of observational errors. More generally, by examining the difference, $\Delta SL^{\dagger} - w$, we can gain insight into biases associated the method (*cf.* Backus 1970; Parker 1994).

To illustrate these ideas, we show in Fig. 6(a) such an averaging function targeted at West Antarctica. Specifically, we have used a Gaussian function as defined as in eq. (32) of Wahr *et al.* (1998), with its centre at (82°S, 110°W), and with a half-width of 500 km. Strictly, this averaging function is not identically zero within the oceans, but it has been checked that this does not affect the results appreciably. A value of −0.0823 m as an equivalent water thickness was determined for the GRACE estimate from our synthetic data set assuming a truncation-degree of $L = 100$. This compares to the true value of −0.0877 m, meaning a relative error of −6.23 per cent associated with the method. To understand this difference, we show in Fig. 6(b) the sensitivity kernel with respect to direct load for the estimate and in (c) the relative difference between this kernel and the averaging function (with values normalized relative to the maximum value of the latter field). Although the relative difference between w and $\Delta SL^{\dagger}$ is everywhere below 1 per cent, this difference is non-zero over much of the Earth's surface, and it is this that accounts for the comparatively large error in the GRACE estimate. Analogous calculations are shown in Fig. 7 for an averaging function targeted on the Greenland ice sheet; in detail the function is centred on (73°N, 40°W) and has a half-width of 400 km. In this case, the relative error in the GRACE estimate was found to be 2.98 per cent which is, again, significantly larger than the point-wise difference between the averaging function and associated sensitivity kernel. The forward-modelling results discussed here are consistent with those of Sterenborg *et al.* (2013), but what is new is the use of sensitivity kernels to understand the source of the bias.

A point worth considering is that, while GRACE does not provide coefficients for $l \leq 1$, the averaging function for the region of interest will usually have some power at these degrees. It has been shown (e.g. Chambers *et al.* 2007) that this can lead to significant biases within the load estimates, while efforts to constrain the missing coefficients have been discussed using additional information

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(a) West Greenland averaging function

(b) Load sensitivity kernel

(c) Difference between kernel and averaging function

(d) Difference between kernel and averaging function with $l \leq 1$ removed

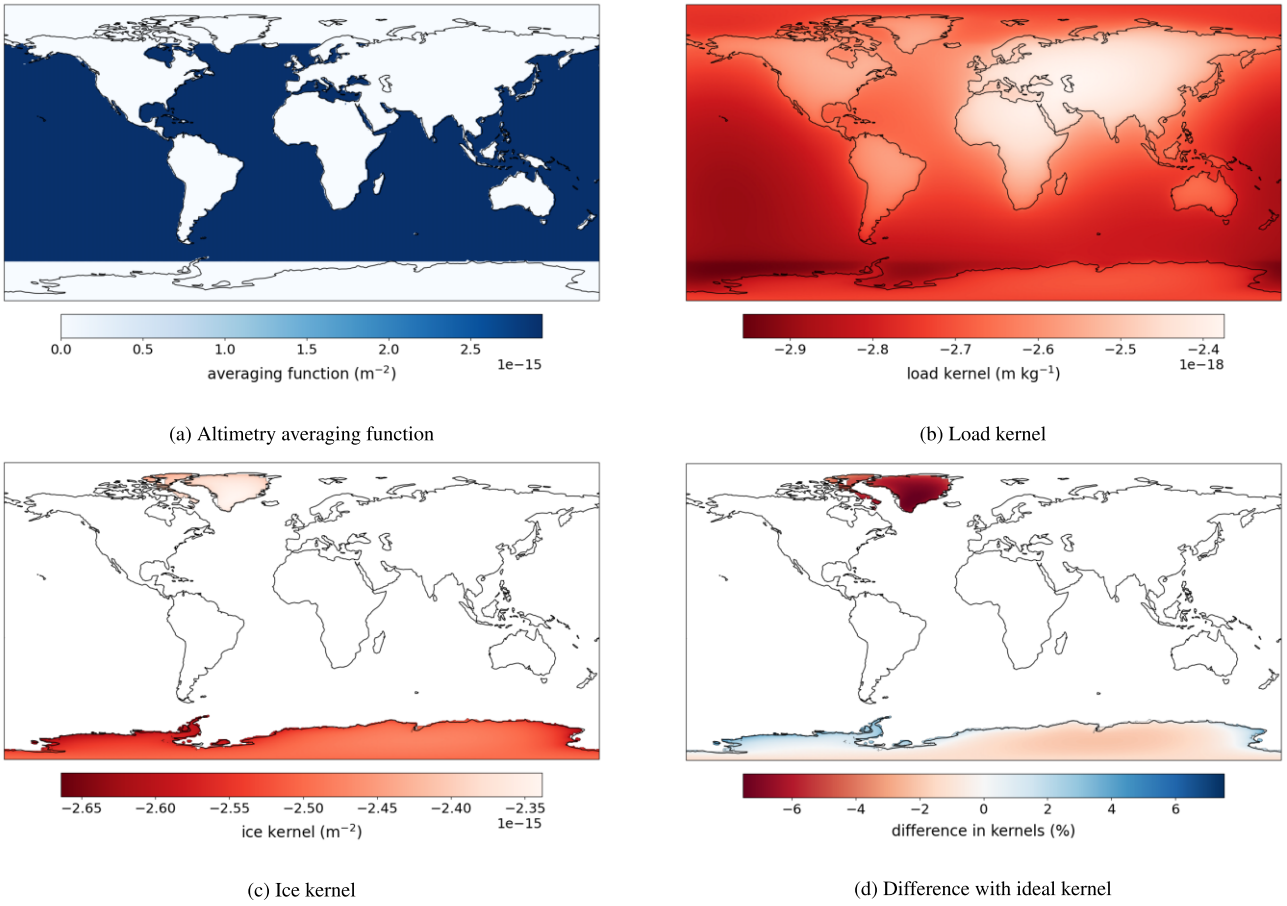
Figure 7. As for Fig. 6 but for an averaging function targeted at Greenland with a half-width of 400 km.

and assumptions (Chambers *et al.* 2004; Swenson *et al.* 2008; Sun *et al.* 2016). It might, therefore, be reasonably asked if the errors seen within our GRACE estimates are due predominantly to missing data at degrees $l \leq 1$. Indeed, within panel (c) of both Figs 6 and 7(a) pronounced degree-one signal is visible. To investigate this issue quantitatively, we can repeat the above calculations but with the degrees $l \leq 1$ removed from the averaging functions. Doing this does not, of course, provide a solution to the degree-one problem, but it allows us to isolate the effect of induced water loading. In the West Antarctic case, the modified calculations lead to 5.37 per cent difference between the GRACE estimate and the true value (meaning $\int_{\partial M} w\,\zeta\,\mathrm{d}S$ determined directly from ζ using the new averaging function), while for Greenland the difference was 1.81 per cent. Within panel (d) of Figs 6 and 7 we show the corresponding differences between load kernels and averaging functions. The dominant degree-one patterns have been removed, and average amplitudes are reduced by a factor of around two, but there remain significant differences over much of the Earth's surface. Moreover, we can see more clearly high-amplitude features within the oceans close to the targeted locations. This makes sense, because this is where sea level will be most strongly affected by loads within the averaging region. These results suggest that (i) missing data for degrees $l \leq 1$ and (ii) the neglect of induced water loading lead to similar magnitude errors within GRACE estimates. As noted by Sterenborg *et al.* (2013), while such errors may be small relative to other sources of uncertainty (e.g. van der Wal *et al.* 2015) they are not negligible, and because they are systematic they should be addressed if at all possible.

Before concluding this example two further comments are worth making. First, within the traditional theory for estimating surface loads from GRACE data (e.g. Wahr *et al.* 1998; Swenson & Wahr 2002), a direct load at degree-one is associated with a degree-one gravitational potential perturbation that cannot be observed. A degree-one direct load would, however, induce a water load that has power at higher degrees and so *is* associated with an observable gravitational signal. This suggests that, by more fully accounting for the physics of the problem, it might be possible to constrain degree-one loads without recourse to additional observations or assumptions.

The second point relates to our assumption that the load average is formed over land, this allowing us to replace the total load with the direct load on the left-hand side of eq. (78). Although this step is appropriate for studies focused on ice sheets or hydrology within the continents, GRACE estimates are also used to study mass changes within the oceans. In the latter case, the distinction between the direct load and the total load is more subtle. We will not enter deeply into this issue, but note related work on the importance of self-gravitation and loading (SAL) within ocean dynamics (e.g. Tamisiea *et al.* 2010; Vinogradova *et al.* 2010, 2015). Such studies have shown, first, that SAL has a significant effect on observable properties such as sea surface height, and secondly, that for timescales of more than around one week SAL effects within ocean simulations can be well-approximated using a *post hoc* hydrostatic correction based on the sea level equation. Because of the latter property, it is appropriate to consider the gravitational signal of ocean dynamic as comprising the response to a direct dynamic

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(a) Altimetry averaging function

(b) Load kernel

(c) Ice kernel

(d) Difference with ideal kernel

Figure 8. In (a), we show the averaging function used to obtain an estimate of $\Delta GMSL$ from sea surface height observations. This function is equal to a constant in the open oceans between $\pm 66^\circ$ in latitude, zero elsewhere, and integrates to one over the Earth's surface. The sensitivity kernel for such a measurement with respect to direct load is shown in (b) and with respect to ice thickness within glaciated regions in (c). The relative differences between the kernel in (c) and the kernel that would return the exact value of $\Delta GMSL$ is shown in (d), with values normalized relative to the latter kernel. Regions in (d) where the difference is positive will have their contributions to the $\Delta GMSL$ estimate artificially amplified, and conversely where the difference is negative.

load along with a gravitationally self-consistent induced load. The method of Wahr *et al.* (1998) and Swenson & Wahr (2002) provides estimates of the total load, but does not, in an obvious manner, allow for the underlying dynamic load to be isolated. This point should be borne in mind when comparing observations to modelling results that have not accounted for SAL. More generally, by adapting the methods discussed above, the relation between GRACE observations and the dynamic component of ocean loads could be quantified.

4.6 Sensitivity kernels for sea surface height change

We now consider, in an idealized way, how global mean sea level change, $\Delta GMSL$, might be estimated from satellite altimetry data (e.g. Nerem *et al.* 2010; Ablain *et al.* 2015). Suppose that we have measurements of sea surface height change, ΔSSH, within the open oceans between $\pm 66^\circ$ in latitude. An estimate of $\Delta GMSL$ can then be obtained by forming the average

$$\Delta GMSL \approx \int_{\partial M} w\, \Delta SSH \, \mathrm{d}S, \tag{80}$$

where w is a function that has support limited to regions where data is available; we will call the right-hand side of this expression an 'altimetry estimate'. Following Ablain *et al.* (2015), the averaging function we use is a constant within the open oceans between latitudes $\pm 66^\circ$, zero elsewhere, and such that $\int_{\partial M} w \, \mathrm{d}S = 1$; note that their 'cosine of latitude weighting' is provided by the surface element in the integral. To proceed, we recall (e.g. Lickley *et al.* 2018) that the sea surface height change can be expressed in the form

$$\Delta SSH = \Delta SL + u + \frac{\psi}{g}. \tag{81}$$

Note that here we are choosing to remove the centrifugal contribution to the sea surface height change. Whether and how this should be done is a subtle point related to reference frames that is discussed in detail by Tamisiea (2011) and Lickley *et al.* (2018). For these illustrative purposes, however, this choice does not matter appreciably. The altimetry averaging method was applied to the synthetic data set shown in Fig. 1, leading to an estimate of 4.90 mm for global mean sea level change, this differing by -1.99 per cent from the true value of 5.00 mm. By repeating such calculations for different melt geometries, it is possible to quantify biases associated with the method. For example, a second synthetic data set was generated in which melting was limited to the northern hemisphere and, in this case, the relative error was larger at -6.74 per cent. Although such a forward modelling approach is undoubtedly useful, it does not provide a clear means for understanding the source

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of the systematic bias and hence is of limited utility in attempts to mitigate against it.

Following the now familiar method, we can apply eq. (57) to obtain

$$\int_{\partial M} w\, \Delta SSH \, \mathrm{d}S = \int_{\partial M} \Delta SL^{\dagger}\, \zeta \, \mathrm{d}S, \tag{82}$$

where $\Delta SL^{\dagger}$ is obtained through solution of the generalized fingerprint problem subject to loads

$$\zeta^{\dagger} = w, \quad \zeta_u^{\dagger}(\mathbf{x}) = -w, \quad \zeta_{\phi}^{\dagger} = 0, \quad \mathbf{k}^{\dagger} = \int_{\partial M} \zeta^{\dagger}\, \mathbf{x} \times (\boldsymbol{\Omega} \times \mathbf{x})\, \mathrm{d}S. \tag{83}$$

Within Fig. 8, we show the results of such a calculation for the constant averaging function discussed above. Supposing that the sea level change is driven entirely through mass transfer from the ice sheets, we can write

$$\Delta GMSL = -\frac{\rho_{\mathrm{i}}}{\rho_{\mathrm{w}} A} \int_{\partial M} (1 - C)\, \Delta I \, \mathrm{d}S. \tag{84}$$

It follows that the altimetry estimate would provide an exact measure of $\Delta GMSL$ if its sensitivity kernel with respect to ice thickness in glaciated regions was equal the constant $-\frac{\rho_{\mathrm{i}}}{\rho_{\mathrm{w}} A}$. From panel (c) within Fig. 8, we can see, however, that this kernel undergoes spatial variations of order of 10 per cent, with its value in Greenland being uniformly lower than in Antarctica. To understand the implications more clearly, we show in panel (d) the relative difference between the ice kernel and the idealized kernel defined through eq. (84). Within the northern hemisphere the difference is largely uniform with value around -6 per cent. In Antarctica, however, the difference between the kernels is not of a uniform sign, with values of around $+2$ per cent found in the west of the continent and mostly negative ones around -2 per cent in the east. These properties suggest that the altimetry averaging method systematically underestimates northern hemisphere contributions to global mean sea level, with this bias evident within the forward calculations discussed earlier within this subsection.

5 DISCUSSION

Within this paper we have established a number of reciprocity theorems related to surface loading and sea level change within an elastic earth model. This provides a simple method for deriving and calculating sensitivity kernels for sea level and related observables with respect to changes in ice thickness. Although we have considered a range of different observables, these examples are not exhaustive. Using the techniques developed it would be relatively easy to obtain kernels for other relevant quantities such as absolute gravity (e.g. van Dam *et al.* 2017), ice surface altimetry (e.g. Rémy & Parouty 2009) or horizontal component GPS measurements (e.g. Coulson *et al.* 2021). Our methods, moreover, apply directly to other kinds of surface loading, such as that associated with ocean dynamics, hydrological processes or sedimentation.

A range of example calculations have been presented to demonstrate the validity of the theoretical results and to point to some potential applications. In future work, we will show that these ideas, when combined with modern geophysical inverse theory, can be used to obtain better constraints on global mean sea level and ice sheet mass loss. To give an indication of how this might be done, we have seen that estimates of global mean sea level change obtained from uniform spatial averages of satellite altimetry data suffer from systematic biases. It is, however, likely that some non-uniform average of the altimetry data will do a better job. For any proposed average, we can use the tools developed in this paper to calculate the associated sensitivity kernel, and hence quantify its performance as an estimator for global mean sea level change. In this manner, the design of optimal estimators for specific quantities of interest can be undertaken following well-established theoretical methods (e.g. Backus & Gilbert 1968; Backus 1970; Stark 2008; Stuart 2010). Here, of course, it would be necessary to account for the full complexity of the real-world problems including data errors and the need to correct, albeit imperfectly, for physical process such as glacial isostatic adjustment and ocean dynamics that affect the observations (e.g. Chambers *et al.* 2010; Nerem *et al.* 2010; Hay *et al.* 2015; Kopp *et al.* 2015; Horton *et al.* 2018).

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DATA AVAILABILITY STATEMENT

Codes needed to reproduce all calculations and figures can be found at https://github.com/da380/SLReciprocityGJI.git.

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